

DSA 20 PATTERNS CHEAT SHEET

500 random problems nahi. Ye 20 pehchaan le — har naya question inhi mein se kisi ek ka roop hai.

@iamxdev

KAISE USE KARE

1. **Solve karne se pehle** 10 sec ruk — guess kar ye kaunsa pattern hai. Galat bhi hua toh dimaag pakadna seekhega.
2. **Solve ke baad** problem ka naam nahi, **pattern ka naam** likh.
3. Ek pattern utha, uske **3-5 problem ek saath** kar — taaki alag dikhne wale same andar se same lagein.

01 Two Pointers

PEHCHAAN Sorted array + ek pair/triplet dhundo, ya dono end se compare (palindrome, reverse).

PRACTICE

- Two Sum II (sorted)
- 3Sum
- Container With Most Water

02 Sliding Window

PEHCHAAN Longest / shortest / max / min contiguous subarray ya substring jo koi condition meet kare.

PRACTICE

- Max Sum Subarray of Size K
- Longest Substring Without Repeating Characters
- Minimum Window Substring

03 Fast & Slow Pointers

PEHCHAAN Linked list mein cycle, middle node, ya 'happy number' type repeat.

PRACTICE

- Linked List Cycle
- Middle of the Linked List
- Happy Number

04 Prefix Sum

PEHCHAAN Baar-baar range-sum, ya 'subarray sum = k' / count subarrays.

PRACTICE

- Subarray Sum Equals K
- Range Sum Query - Immutable
- Find Pivot Index

05 Hashing / Frequency Map

PEHCHAAN Count karna, fast lookup, duplicate ya anagram, 'isko pehle dekha kya?'

PRACTICE

- Two Sum
- Group Anagrams
- Longest Consecutive Sequence

06 Binary Search

PEHCHAAN Sorted + search, YA 'minimize the max / maximize the min' (answer pe search).

PRACTICE

- Search in Rotated Sorted Array
- Koko Eating Bananas
- Median of Two Sorted Arrays

07 Merge Intervals

PEHCHAAN Intervals / start-end time / overlap / merge / meeting rooms.

PRACTICE

- Merge Intervals
- Insert Interval
- Meeting Rooms II

08 Cyclic Sort

PEHCHAAN Array mein n numbers range [1..n] ke, missing ya duplicate dhundo.

PRACTICE

- Missing Number
- Find All Numbers Disappeared in an Array
- First Missing Positive

09 In-place LL Reversal

PEHCHAAN Poori linked list ya uska ek part reverse karna.

PRACTICE

- Reverse Linked List
- Reverse Nodes in k-Group
- Reorder List

10 Monotonic Stack

PEHCHAAN Next / previous greater ya smaller element, histogram, span.

PRACTICE

- Next Greater Element I
- Daily Temperatures
- Largest Rectangle in Histogram

11 Top-K Elements (Heap)

PEHCHAAN Top / smallest / largest / 'K most frequent' nikaalna.

PRACTICE

- Kth Largest Element in an Array
- Top K Frequent Elements
- K Closest Points to Origin

12 Two Heaps

PEHCHAAN Running median, ya do halves ko balance karna.

PRACTICE

- Find Median from Data Stream
- Sliding Window Median
- IPO

13 Tree BFS (Level Order)

PEHCHAAN Level-by-level chalo, 'level order', min depth, right side view.

PRACTICE

- Binary Tree Level Order Traversal
- Zigzag Level Order Traversal
- Binary Tree Right Side View

14 Tree DFS

PEHCHAAN Root-to-leaf path, validate BST, height / diameter, LCA.

PRACTICE

- Maximum Depth of Binary Tree
- Validate Binary Search Tree
- Lowest Common Ancestor of a BST

15 Graph BFS / DFS

PEHCHAAN Grid / islands, connected components, flood fill, unweighted shortest path.

PRACTICE

- Number of Islands
- Rotting Oranges
- Clone Graph

16 Topological Sort

PEHCHAAN Dependencies ka order, prerequisites, build order.

PRACTICE

- Course Schedule
- Course Schedule II
- Alien Dictionary

17 Union-Find (DSU)

PEHCHAAN Connectivity, groups, undirected graph mein cycle ya redundant edge.

PRACTICE

- Number of Provinces
- Redundant Connection
- Accounts Merge

18 Backtracking

PEHCHAAN 'Saare' combinations / permutations / subsets generate karo, ya constraints (N-Queens, Sudoku).

PRACTICE

- Subsets
- Permutations
- Combination Sum

19 Dynamic Programming

PEHCHAAN Count ways / min-max cost / 'can you reach' + overlapping subproblems + choices.

PRACTICE

- Climbing Stairs
- House Robber
- Coin Change

20 Greedy

PEHCHAAN Har step pe locally best pick (jumps, intervals, activity selection).

PRACTICE

- Jump Game
- Non-overlapping Intervals
- Gas Station

21 Bit Manipulation

PEHCHAAN Single number, XOR tricks, subsets via bitmask, count bits.

PRACTICE

- Single Number
- Counting Bits
- Subsets (bitmask)

Ginti nahi. Pattern.

Ye sheet sirf shuruaat hai. Core CS theory, OA + aptitude, aur resume sheets aa rahi hain.
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